TTAGOT Reasoning System Benchmark Results

MCP integration and adaptive reasoning evaluated on GPQA Diamond

TTAGOT explores structured reasoning for complex questions through the Model Context Protocol (MCP). It combines an independent direct-analysis path with TTAGOT adaptive graph reasoning, then compares candidate answers and their supporting evidence.

Engineering contribution

The project brings task planning, branch evaluation and final answer review into an MCP workflow. Its engineering work combines adaptive graph orchestration with independent analysis and compares six models using benchmark accuracy, uncertainty and API cost.

How TTAGOT works

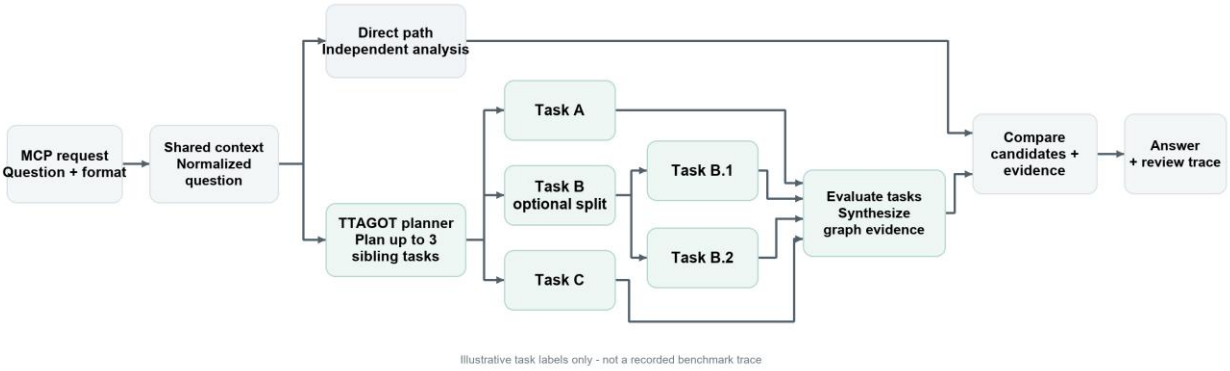


Figure 1. An illustrative branching case shows sibling tasks and a nested decomposition. The graph labels are examples, not a recorded benchmark trace.

Results at a glance

Comparison	Outcome in this sample	TTAGOT difference
No-reasoning baseline	TTAGOT scored higher in 5 of 5 comparisons	+29.2 to +38.9 pp
Provider reasoning: Gemini	TTAGOT scored higher in 3 of 3	+1.5 to +7.6 pp
Provider reasoning: GPT-5	Provider reasoning scored higher in all 3	-9.5 to -5.6 pp

Gemini 2.5 Pro + TTAGOT had the highest observed accuracy at 89.4% and cost $0.2631 per question. Gemini 2.5 Flash + TTAGOT reached 84.8% at $0.0297.

Accuracy by reasoning mode

The benchmark covered 198 graduate-level STEM multiple-choice questions, with one evaluation per question in each available condition. No reasoning is the reported baseline without provider reasoning; provider reasoning uses the provider's own mode. TTAGOT adds the graph workflow. Provider reasoning remained enabled for Gemini 2.5 Pro, which has no no-reasoning baseline.

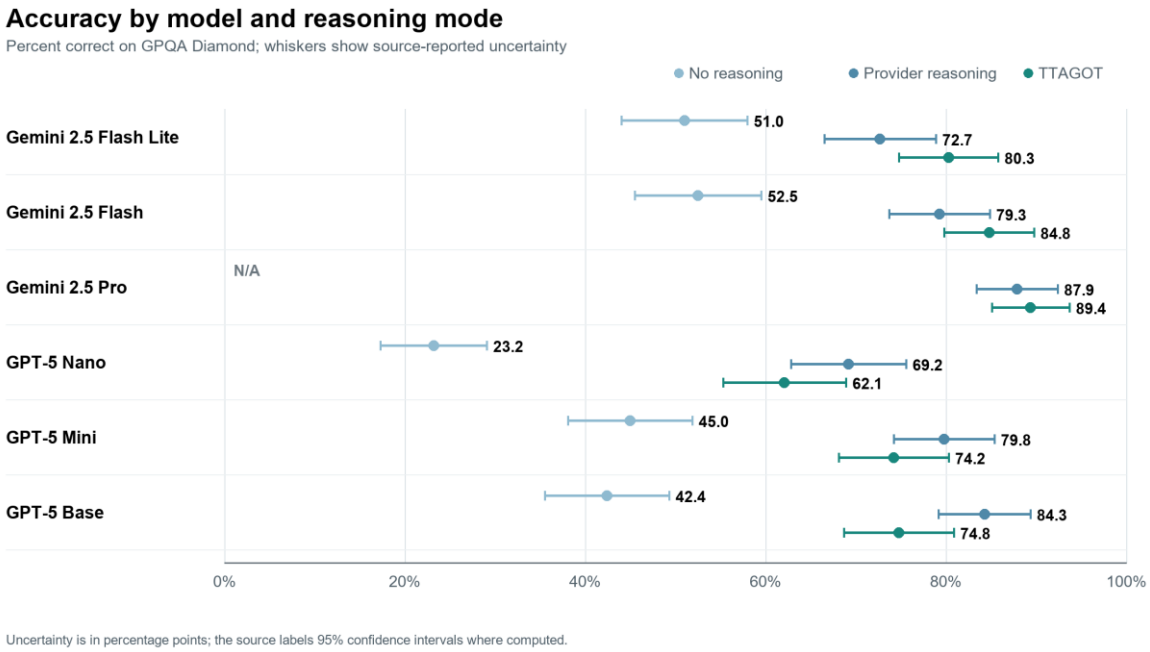


Figure 2. Mean accuracy and source-reported uncertainty. Uncertainty is shown in percentage points; the source reports 95% confidence intervals where computed.

Model	No reasoning	Provider reasoning	TTAGOT	TTAGOT vs no reasoning	TTAGOT vs provider
Gemini 2.5 Flash Lite	51.0 +/- 7.0	72.7 +/- 6.2	80.3 +/- 5.5	+29.3 pp	+7.6 pp
Gemini 2.5 Flash	52.5 +/- 7.0	79.3 +/- 5.6	84.8 +/- 5.0	+32.3 pp	+5.5 pp
Gemini 2.5 Pro	N/A	87.9 +/- 4.5	89.4 +/- 4.3	N/A	+1.5 pp
GPT-5 Nano	23.2 +/- 5.9	69.2 +/- 6.4	62.1 +/- 6.8	+38.9 pp	-7.1 pp
GPT-5 Mini	45.0 +/- 6.9	79.8 +/- 5.6	74.2 +/- 6.1	+29.2 pp	-5.6 pp
GPT-5 Base	42.4 +/- 6.9	84.3 +/- 5.1	74.8 +/- 6.1	+32.4 pp	-9.5 pp

Accuracy and uncertainty are percentages; differences are percentage points (pp). N/A denotes an unavailable baseline. GPT-5 Base TTAGOT accuracy follows Table 4.2 (74.8%).

Source: Advanced Reasoning MCP, Chapters 1-6, Chapter 4, Table 4.2.

Cost and accuracy

These are historical study averages for TTAGOT accuracy, token use and API cost per question. The comparison below pairs Gemini 2.5 Pro with provider reasoning alone against Gemini 2.5 Flash with TTAGOT.

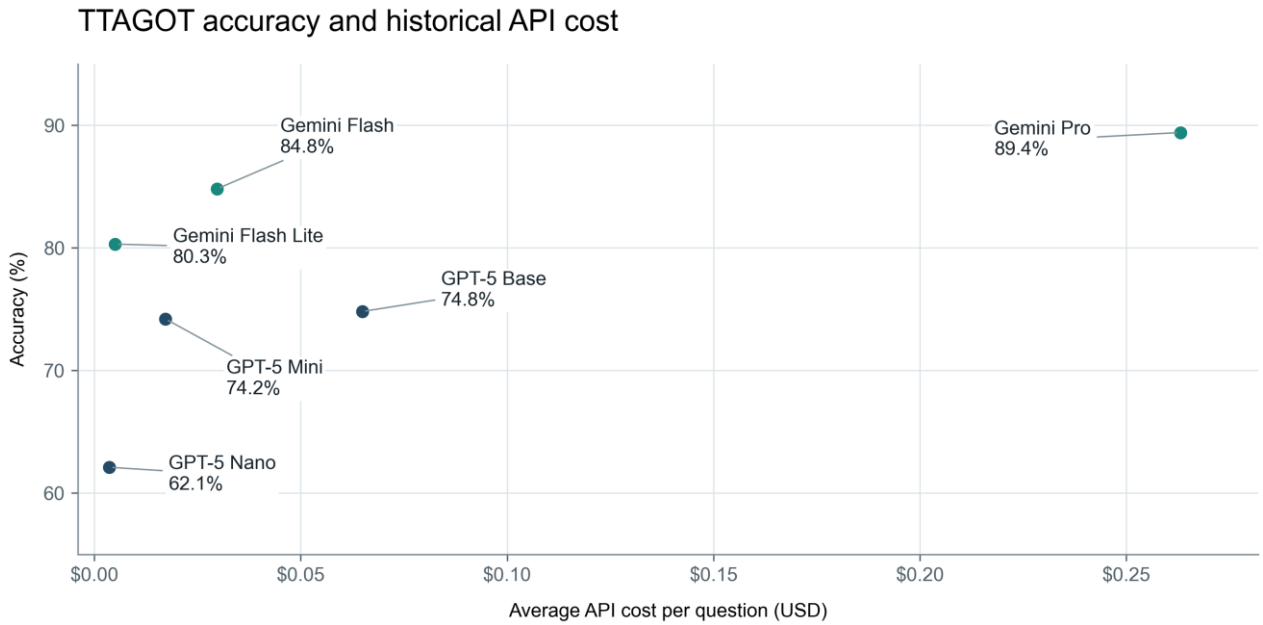


Figure 3. Historical API cost and accuracy for the six TTAGOT configurations. Source: Chapter 4, Table 4.4; GPT-5 Base accuracy follows Table 4.2.

Configuration	Accuracy	Tokens per question	Cost per question
Gemini 2.5 Flash Lite + TTAGOT	80.3%	24,924	$0.0050
Gemini 2.5 Flash + TTAGOT	84.8%	30,867	$0.0297
Gemini 2.5 Pro + TTAGOT	89.4%	34,508	$0.2631
GPT-5 Nano + TTAGOT	62.1%	27,597	$0.0036
GPT-5 Mini + TTAGOT	74.2%	25,898	$0.0172
GPT-5 Base + TTAGOT	74.8%	22,149	$0.0649

Gemini Pro and Flash comparison

Condition	GPQA Diamond accuracy	Cost per question
Gemini 2.5 Pro without TTAGOT (provider reasoning)	87.9%	$0.0729
Gemini 2.5 Flash with TTAGOT	84.8%	$0.0297

Flash with TTAGOT cost 59.3% less per question than Pro with provider reasoning alone, with observed accuracy 3.1 percentage points lower.

Evaluation notes

One evaluation was run per question and condition. The source reports uncertainty intervals, with no repeated trials, hypothesis tests or effect sizes. These results cover graduate-level STEM multiple-choice questions.

Source: Advanced Reasoning MCP, Chapters 1-6, Chapter 4, Tables 4.2-4.4 and cost discussion.